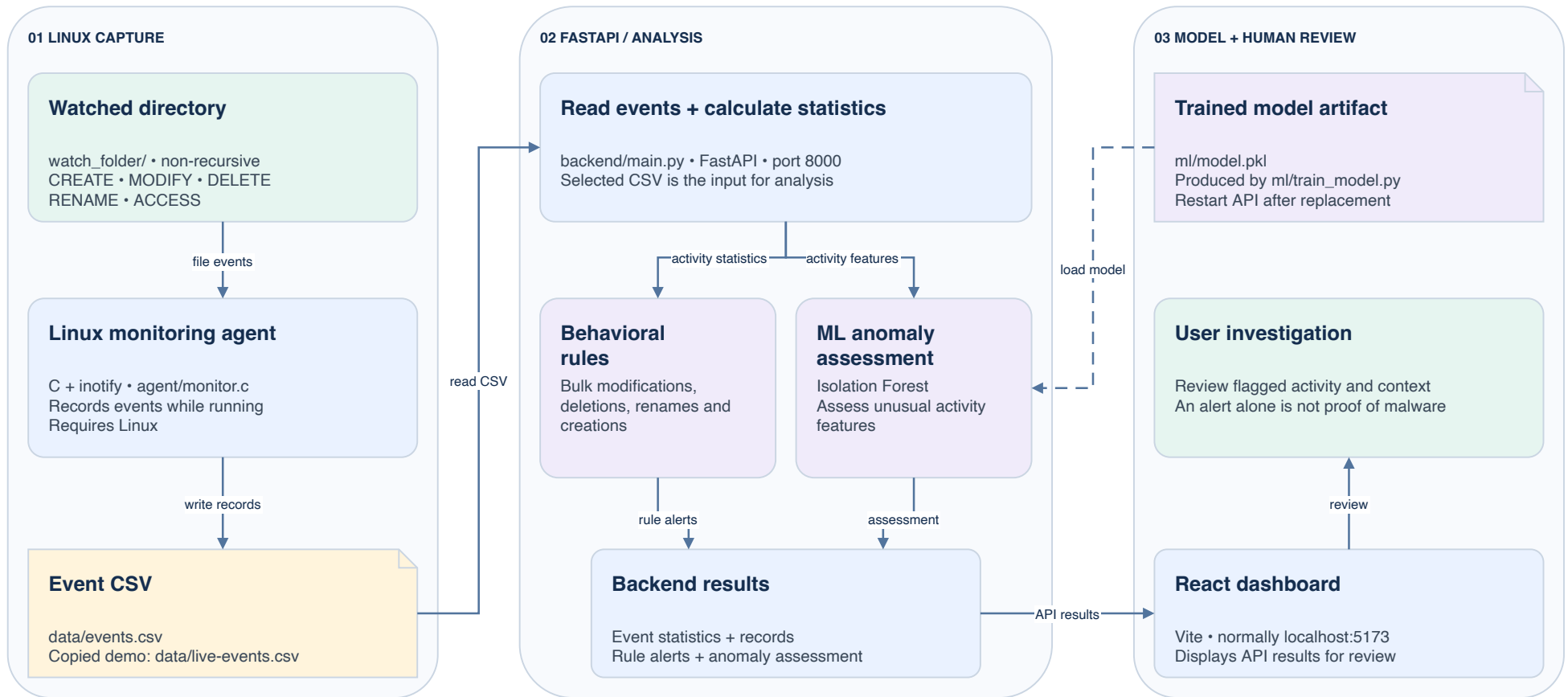


FileSentinel AI | Architecture overview

File-system telemetry → behavioral analysis → visible evidence → human investigation



SOLID: event / result flow • DASHED: offline artifact dependency • FOLDED CORNER: persisted file

Cross-environment boundary: the verified Linux VM → Mac demo requires manual CSV transfer. New VM events need another transfer.
Alternative input: scripts/test_simulation.py appends synthetic records to data/events.csv; it does not perform real file operations.

The event-to-alert workflow

Numbered logical flow • implementation details not supplied are deliberately left unspecified

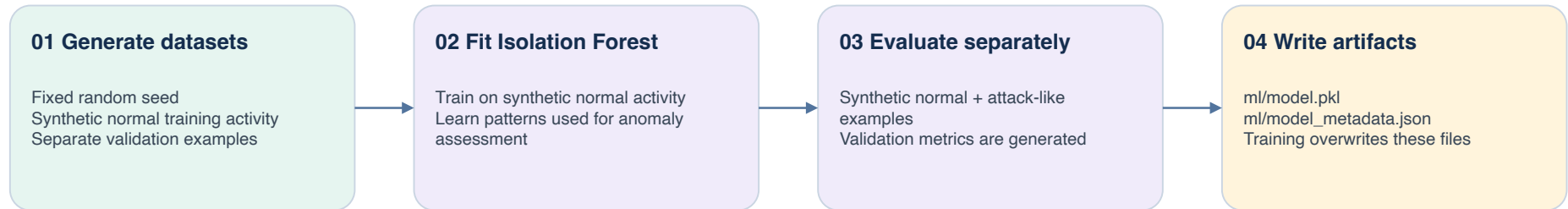


Contract detail: exact CSV columns, feature formulas, API routes, rule thresholds and refresh cadence require source-code verification.

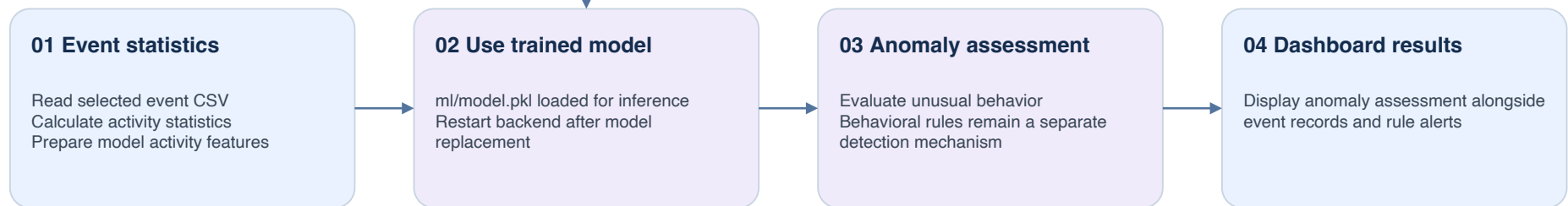
Training and inference are separate

Isolation Forest • reproducible synthetic training • explicit model artifacts • measured claims

OFFLINE / ml/train_model.py



RUNTIME / FastAPI backend

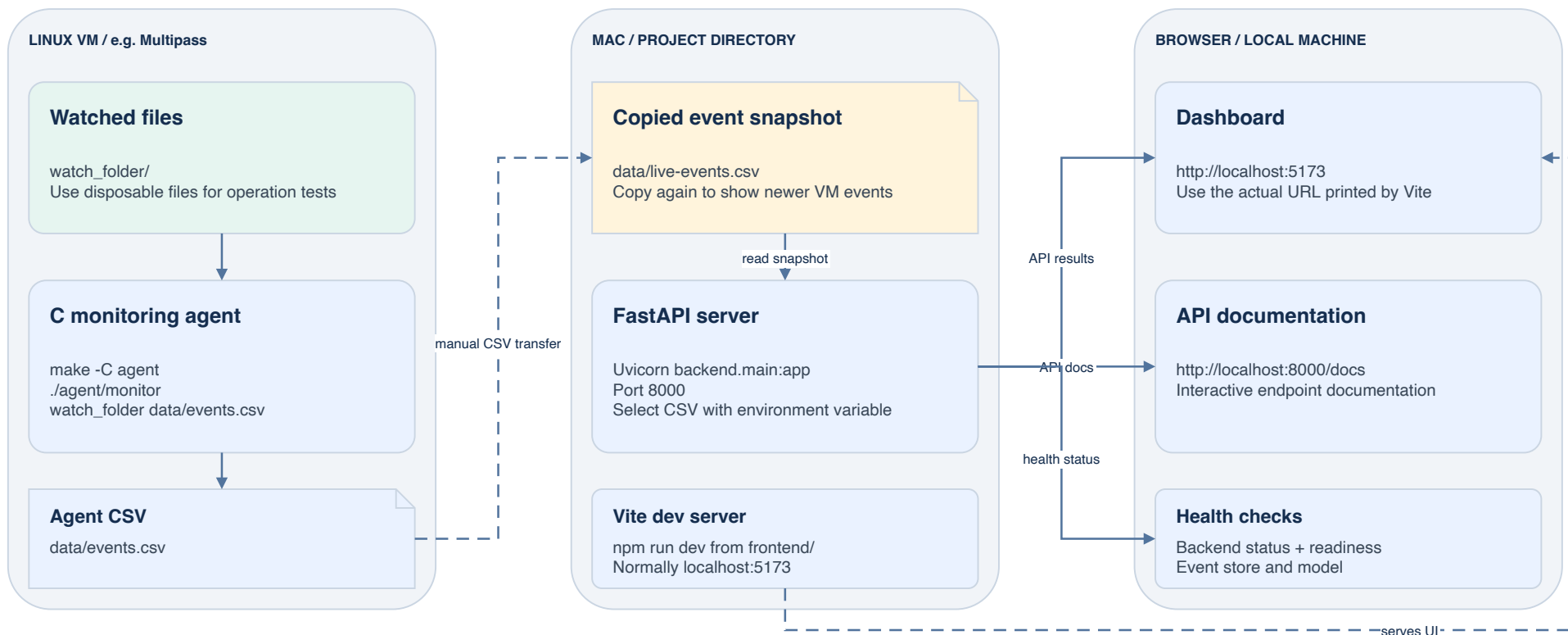


How to explain the ML results accurately

Synthetic validation measures performance on generated examples, not established real-world detection accuracy. Review ml/model_metadata.json for actual metrics. Dashboard risk scores are not model accuracy or validated infection probabilities. Exact feature definitions and score transformations were not provided.

Linux capture, Mac analysis, browser review

Documented demonstration topology • manual snapshot transfer is the key environment boundary



Backend input selection

Set FILESENTINEL_EVENTS_FILE to the absolute path of data/live-events.csv before starting the backend. Restart without that override to return to the standard data/events.csv workflow. The Vite server serves the UI; the browser uses the FastAPI service for data. The agent must run on Linux.

Code ownership and evidence map

A project map for reviewers • current documentation takes priority over the older folder outline

CAPTURE / agent/

monitor.c • Makefile • README.md
Linux inotify monitor, build process and agent setup instructions.

SERVICE / backend/

main.py • requirements.txt
API, event processing and detection logic. Internal module names in the outline need code verification.

INTERFACE / frontend/

React dashboard + Vite
Supplied lockfile: dashboard 2.0.0; React/react-dom 19.2.7; Vite 8.1.0; React plugin 6.0.3.

DATA + MODEL / data/ + ml/

events.csv • live-events.csv demo
train_model.py • model.pkl
model_metadata.json
Events, training and persisted artifacts.

UTILITIES + TESTS

scripts/test_simulation.py
Automated tests/ • pytest
Synthetic records exercise the pipeline. README records 12 local tests passed on October 1, 2026.

DOCUMENTATION / docs/

README.md
docs/images-filesentinel/
Dashboard, API, health and Linux event screenshots document observed behavior.

DOCUMENTED LOCAL EVIDENCE

Agent compiled in Linux • CREATE and MODIFY captured • Two copied Linux events displayed •
Simulation produced four rule alerts • 12 tests passed in the recorded run.

VERIFY BEFORE CLAIMING

GitHub Actions status and current training metrics require separate checks. SQLite/database modules and Docker Compose appear in the older outline but are not established as active by the README. Lockfile versions do not prove runtime installation.